Cosmology as a Synchronous Specular Bit Architecture Aligned with SBA–DNA Formalism

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Abstract—This manuscript presents a coherent cosmological interpretation framework grounded in the formal Specular Bit Architecture (SBA). The central contribution is an evidentiary separation between formal claims (proved), modeling claims (derived), and physical claims (hypothesis-level). The formal layer uses specular signed carriers, exact local Fibonacci rewrites, deterministic sanitization, and Protocol C arbitration. The modeling layer defines dimensional bridges and calibration constraints. The interpretation layer is explicitly falsifiable through pre-registered observables and uncertainty-qualified tests.

Beyond this separation, the manuscript contributes a structured verification pipeline: (i) local algebraic invariants are stated in machine-checkable form, (ii) synchronous update semantics are linked to deterministic replay and bounded commit magnitude, (iii) calibration maps are framed as identifiable statistical objects with uncertainty propagation, and (iv) hypothesis-level conclusions are constrained by pre-declared rejection rules. This organization is intended to prevent category errors in which mathematical consistency is misread as empirical confirmation.

By enforcing these boundaries, the manuscript improves logical consistency, mathematical traceability, and scientific rigor while preserving cross-domain relevance to arithmetic theory, substrate engineering, and cosmological hypothesis design. The resulting document is designed so that reviewers can independently accept or reject formal, modeling, and interpretive layers without collapsing the entire inference chain.

Index Terms—Specular Bit Architecture, Protocol C, deterministic sanitization, Fibonacci positional weights, informational mass–energy, DNA storage, neuromorphic photonics, falsifiability.

1 Introduction

The principal objective is to retain conceptual scope without conflating proof, calibration, and interpretation. Many interdisciplinary proposals fail because formal correctness is treated as physical confirmation. This manuscript avoids that failure mode by defining inference boundaries before presenting results.

Claims are partitioned as follows:

- **Formal layer:** propositions derivable from carrier algebra, rewrite identities, and deterministic synchronous execution.
- **Model layer:** dimensional relations that introduce calibration constants and uncertainty assumptions.
- **Hypothesis layer:** cosmological interpretation statements requiring independent empirical falsification.

This structure allows reviewers to evaluate each layer independently.

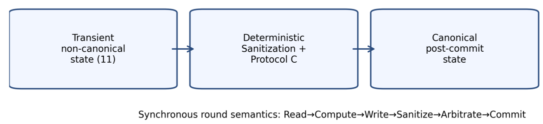

Fig. 1. Deterministic carrier-to-canonical transformation pipeline under synchronous rounds.

This manuscript is accompanied by supplemental material in `SBA_DNA.pdf`, which provides aligned SBA–DNA companion context for the interpretation layer.

To make this partition operational, we define a one-way admissibility rule:

$$\mathcal{E}_{\mathrm{formal}} \Rightarrow \mathcal{E}_{\mathrm{model}} \Rightarrow \mathcal{E}_{\mathrm{hypothesis}},$$

where each implication indicates *permission to build upon*, not logical equivalence. In particular, no amount of hypothesis-level fit can retroactively validate a broken formal invariant.

A practical consequence is that every major claim should be traceable to a tuple

$$\Gamma = (\mathcal{A}, \mathcal{L}, \mathcal{D}, \mathcal{F}),$$

where $\mathcal{A}$ is the assumption set, $\mathcal{L}$ is the layer tag, $\mathcal{D}$ are declared dependencies, and $\mathcal{F}$ is the falsification condition. This explicit tuple improves auditability and supports modular peer-review.

2 Interpretive Architecture and Evidence Flow

A fixed three-layer mapping is used:

1) transient non-canonical microstates (probability/acausal layer),
2) synchronous update cadence (light/clock layer),
3) post-commit canonical occupancy (structured/mass layer).

Figures 1 and 2 define admissible claim flow in this manuscript.

Remark 1 (Mirror-geometry interpretation layer)**.** *In the bridge interpretation aligned with the SBA–DNA companion text*

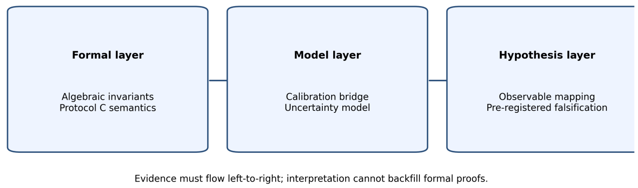

Fig. 2. Evidence flow rule: formal $\rightarrow$ modeling $\rightarrow$ hypothesis. Reverse inference is disallowed.

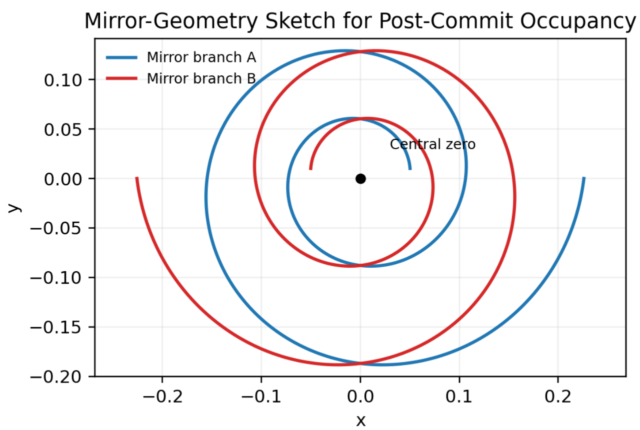

Fig. 3. Conceptual local mirror architecture around a central axis ("central zero" in the model vocabulary), with two specular branches (interpretation layer).

TABLE 1
Bridge between post-commit occupancy and SBA–DNA mirror-geometry interpretation.

Formal/model object	Interpretation-layer geometric reading
Central canonical balance point	Central axis ("central zero" in the model vocabulary)
Dual-channel signed occupancy	Complementary structure represented as specular pairs $01 \leftrightarrow 10$
Commit-stabilized orientation pattern	Strand anti-parallelism interpreted as local mirror architecture
Synchronous round index	Discrete sequencing parameter along mirrored geometry

(Remark 2.2), post-commit canonical occupancy is treated not only as a symbolic endpoint but also as a geometry-generating state. The geometric reading follows the same phrase set used in the SBA–DNA formalism: complementary structure is interpreted as a local mirror architecture around a central axis ("central zero" in the model vocabulary), and complementary microstates are represented as specular pairs $01 \leftrightarrow 10$.

In this interpretation, matter/antimatter language is used only as an analogy for opposite signed informational carriers, not as a biological claim about literal particle content in DNA. Accordingly, this statement remains hypothesis/model-layer language and does not modify formal rewrite correctness, invariance proofs, or Protocol C determinism.

The three-layer map can be interpreted as a constrained information-flow graph. Let X_t denote non-canonical microstate content before sanitization, U_t the synchronous update operator (including arbitration), and C_t the canonical post-commit state. Then

$$C_{t+1} = \Pi_{\text{can}}(U_t(X_t)),$$

where Π_{can} is the canonical projection induced by sanitization and Protocol C commitments. The formal layer studies properties of U_t and Π_{can}; the model layer studies calibrated observables derived from C_t; and the hypothesis layer studies domain interpretations constrained by those observables.

This decomposition clarifies that interpretive narratives are not part of the transition law itself. They are secondary mappings that must preserve layer order and uncertainty accounting.

3 Formal Core: Carrier Semantics and Exact Local Laws

3.1 Notation consistency

Global notation is fixed: δ_i decoded signed state, W_i positional weight, d_i intermediate signed value, $\Sigma(d)$ weighted sum, m_{SBA} canonical occupancy index, and σ specular swap operator.

For clarity, we also define local overflow indicator $\chi_i = \mathbf{1}\{|d_i| = 2\}$, candidate set $\mathcal{K}_j$ for target j, and committed update $v_j \in \{-1, 0, +1\}$. These symbols are used to state deterministic constraints in Sections 4–5.

3.2 Carrier algebra

At index i, poles (p_i, n_i) decode as

$$\delta_i = p_i - n_i \in \{-1, 0, +1\}.$$

Admissible microstates are $00, 01, 10$. State 11 is a transient ambiguity requiring deterministic sanitization.

Specular symmetry:

$$\delta(\sigma(p_i, n_i)) = -\delta(p_i, n_i), \qquad \sigma^2 = \text{Id}.$$

3.3 Fibonacci local rewrite identity

Let $W_i = F_{i+o}$ for fixed offset o. Then

$$2W_i = W_{i+1} + W_{i-2}.$$

Proposition 1 (Local weighted-sum preservation)**.** *Let* $\Sigma(d) = \sum_k d_k W_k$*. For rewrites* $d_i = +2 \rightarrow (d_i - 2, d_{i+1} + 1, d_{i-2} + 1)$ *and* $d_i = -2 \rightarrow (d_i + 2, d_{i+1} - 1, d_{i-2} - 1)$*, one has* $\Delta\Sigma = 0$.

Sketch. For the $+2$ rewrite at index i, the weighted variation is

$$\Delta\Sigma = (-2)W_i + (+1)W_{i+1} + (+1)W_{i-2}.$$

Using $2W_i = W_{i+1} + W_{i-2}$, one obtains $\Delta\Sigma = 0$. The -2 case is symmetric by sign reversal, yielding the same cancellation pattern. $\square$

TABLE 2
Round-phase contract for deterministic replay.

Phase	Input object	Deterministic requirement
Read	latched state	no in-phase mutation of source snapshot
Compute	local candidates	finite stencil and pure-function evaluation
Write	rewrite deltas	simultaneous application from same snapshot
Sanitize	transient flags	canonical projection with explicit policy
Arbitrate	candidate sets	strict total order and reproducible winner
Commit	target updates	per-target value in $\{-1, 0, +1\}$ only

4 Deterministic Synchronous Semantics

One round is

$$\begin{aligned}\text{Read} &\to \text{Compute} \to \text{Write}\\ &\to \text{Sanitize} \to \text{Arbitrate} \to \text{Commit}.\end{aligned}$$

All candidates are computed from one snapshot.

Protocol C imposes deterministic ordering and single-winner commit per target.

Formally, let $\prec$ be the strict total order used by Protocol C over candidates targeting the same index. Arbitration returns

$$v_j = \arg\max_{k \in \mathcal{K}_j}^{\prec} s(k),$$

where $s(k)$ is the deterministic scoring map and ties are broken by $\prec$. If $\mathcal{K}_j = \varnothing$, then $v_j = 0$. This definition makes replay traces platform-independent provided implementations preserve s and $\prec$.

Proposition 2 (Per-target non-accumulation). *With synchronous snapshots and Protocol C ordering, committed update to any target per round belongs to* $\{-1, 0, +1\}$.

Sketch. Because all candidates are derived from one latched snapshot, concurrent writers cannot chain intra-round into unbounded accumulation. Protocol C permits at most one committed winner per target per round, so the resulting committed delta remains bounded in $\{-1, 0, +1\}$. □

5 Termination Measure and Bounded Propagation

Define

$$R(d) = \max\{i : |d_i| = 2\}, \qquad m(d) = |\{i : |d_i| = 2\}|,$$

with lexicographic measure $p(d) = (R(d), m(d))$. Under finite support and local update stencils, $p(d)$ descends across rounds, implying finite normalization by well-foundedness.

This is the formal computational meaning of "classicalization" used here.

A useful quantitative interpretation is to define termination time

$$T(d_0) = \min\{t \geq 0 : |d_i^{(t)}| \leq 1 \ \forall i\}.$$

Under finite support and deterministic rounds, $T(d_0) < \infty$ whenever the lexicographic descent condition is satisfied at each non-canonical step. In practice, reporting empirical upper envelopes for T over benchmark families complements the proof-level argument and helps compare implementations without changing formal semantics.

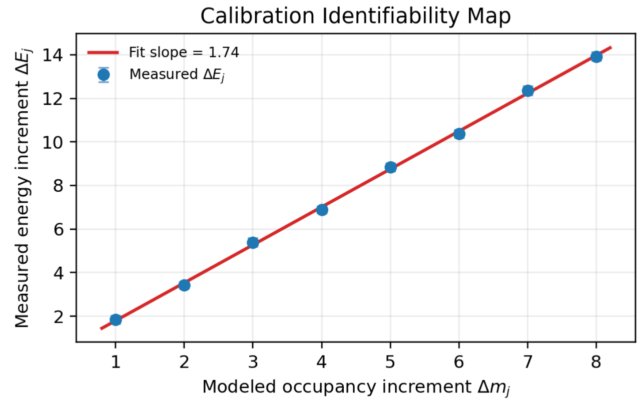

Fig. 4. Illustrative calibration map: measured energy increment versus modeled occupancy increment with uncertainty bars and fitted slope μc^2.

6 Model Layer: Dimensional Bridge and Calibration

Formal occupancy index:

$$m_{\mathrm{SBA}} = \sum_i |a_i| W_i.$$

Dimensional bridge:

$$m_{\mathrm{phys}} = \mu m_{\mathrm{SBA}}, \qquad E = m_{\mathrm{phys}} c^2.$$

Remark 2. *These relations are calibration-dependent mappings, not physical proofs.*

A modeling claim is admissible only if it reports: operation class, Δm_{SBA}, measured ΔE, uncertainty model, and out-of-sample checks.

To avoid non-identifiable calibrations, we recommend estimating μ from multiple disturbance classes with non-collinear Δm_{SBA} support and heteroscedastic uncertainty weighting. A minimal regression template is

$$\Delta E_j = \mu c^2 \Delta m_j + \epsilon_j, \qquad \epsilon_j \sim \mathcal{N}(0, \sigma_j^2),$$

with σ_j reported or estimated from repeated trials. Model adequacy should be checked through posterior predictive residuals and leave-condition-out validation.

7 Hypothesis Layer: Falsifiability Protocol

Hypothesis-level claims must include:

1) explicit formal dependencies,
2) explicit additional assumptions,
3) pre-registered calibration and statistical protocol,
4) pre-declared rejection criteria.

Operationally, each hypothesis must define a pre-analysis plan containing estimator choice, exclusion rules, multiplicity correction (if multiple observables are tested), and stopping criteria. This avoids adaptive analyses that can inflate false positive interpretation claims.

TABLE 3
Validation checklist by evidence layer.

Layer	Required evidence	Failure signal
Formal	exact replay + invariance checks	reproducible non-zero drift
Formal	deterministic arbitration traces	same-target multi-source commit
Modeling	stable posterior with uncertainty bounds	non-identifiable or regime-unstable fit
Hypothesis	independent out-of-sample replication	inconsistent sign or magnitude

TABLE 4
Contrast with adjacent strands.

Strand	Canonical focus	Focus in this manuscript
Signed-digit arithmetic	carry-limited speedups	specular signed carriers with deterministic sanitization
Fibonacci systems	uniqueness and sparsity	local invariant-preserving rewrite laws
Information thermodynamics	erasure-cost constraints	calibration discipline for interpretation limits
Neuromorphic / photonics	substrate acceleration	substrate-agnostic deterministic semantics
DNA data storage	molecular coding robustness	dual-channel compatibility with replay logic

A compact decision functional can be written as

$$\mathcal{D}(H) = \mathbf{1}\{q(H) \le \alpha\} \cdot \mathbf{1}\{\mathrm{sgn}(\hat{\theta}_H) = \mathrm{sgn}(\theta^\star_H)\},$$

where $q(H)$ is the adjusted significance metric and the sign constraint enforces directional consistency with pre-registered expectations.

8 Related Work Contrast

The contrast in Table 4 is not a novelty claim against each strand in isolation; rather, it specifies where this manuscript places its methodological emphasis. The intended contribution is integrative discipline: deterministic local laws, explicit calibration boundaries, and strict falsifiability conditions are presented as a combined workflow.

9 Methodological Program

This section operationalizes the three-layer evidentiary architecture into a concrete research workflow. The intent is to minimize ambiguity about what is being tested at each stage, what constitutes failure, and how uncertainty propagates from low-level implementation artifacts to high-level interpretation claims. Each track has independent deliverables and must pass its own acceptance criteria before downstream use.

9.1 Formal verification track

The formal track evaluates only theorem-level and protocol-level statements. It excludes calibration constants and cosmological language by construction.

9.1.0.1 State-space and transition validation.: For bounded neighborhoods, all admissible local states are enumerated, and one-round transitions are replayed under the synchronous schedule. The primary checks are: (i) weighted-sum preservation under interior and boundary rewrites, (ii) deterministic candidate construction from a single read snapshot, and (iii) per-target non-accumulation after arbitration.

9.1.0.2 Invariant suite.: Each transition instance must satisfy a machine-checkable invariant tuple

$$\mathcal{I} = \big(\Delta\Sigma,\, \Delta\mathcal{N}_{\pm 2},\, \Delta\mathcal{C}_{\mathrm{multi}},\, \Delta\mathcal{R}_{\mathrm{order}}\big),$$

where $\Delta\Sigma = 0$ is required for all rewrite-only updates, $\mathcal{N}_{\pm 2}$ counts overflow sites, $\mathcal{C}_{\mathrm{multi}}$ counts illegal same-target multi-commits, and $\mathcal{R}_{\mathrm{order}}$ encodes arbitration-order consistency. Any non-zero violation in a deterministic replay is a formal failure, not a statistical anomaly.

9.1.0.3 Termination and radius certification.: For finite-support inputs sampled across sparsity and sign configurations, the lexicographic potential $p(d) = (R(d), m(d))$ is tracked round-by-round. Certification requires monotone descent until stabilization and a bounded empirical envelope for propagation radius and round count over the tested regime. The envelope is reported as a function of neighborhood class and boundary type.

9.2 Implementation track

The implementation track measures whether concrete substrates preserve the semantics proved in the formal layer.

9.2.0.1 Cross-platform parity.: Equivalent disturbance corpora are executed on software reference kernels, FPGA prototypes, and photonic emulation pipelines. Platform outputs are aligned by deterministic trace IDs. For each case, parity checks compare event ordering, committed deltas, and final canonical state.

9.2.0.2 Performance with semantic constraints.: Benchmarking reports are normalized by effective update count, not wall-clock alone. Throughput and latency are presented jointly with semantic integrity indicators (e.g., replay agreement, illegal-commit count, invariant drift). A faster implementation is not accepted if semantic mismatch exceeds declared tolerance.

9.2.0.3 Fault-injection protocol.: Controlled perturbations (timing jitter, bit flips, stale reads, and arbitration race pressure) are injected to map semantic robustness margins. The resulting failure surface is used to define safe operating envelopes and to prioritize protocol hardening.

9.3 Molecular mapping track

The molecular track tests whether dual-channel SBA semantics can be preserved under realistic sequence-level constraints.

9.3.0.1 Encoding discipline.: Dual-channel symbols are encoded with explicit index mapping, strand orientation constraints, and sanitization metadata. In the bridge interpretation aligned with Remark 2.2 of the SBA–DNA companion paper, strand orientation is treated as the molecular counterpart of the local mirror architecture around a central axis ("central zero" in the model vocabulary). Concretely, opposite strand directions are mapped as complementary specular pairs $01 \leftrightarrow 10$, preserving deterministic index order and reconstruction constraints. Library preparation includes controls for homopolymer suppression and GC-balance to reduce sequencing bias.

9.3.0.2 Decoding and deterministic reconstruction.: Observed reads are converted to channel candidates under a fixed alignment policy, then reconstructed with the same deterministic arbitration used in digital substrates. The key metric is reconstruction determinism under repeated sequencing runs and varying noise profiles.

9.3.0.3 Noise-aware validation.: Error classes (substitution, insertion, deletion, dropout) are modeled separately. Validation reports must include per-class confusion matrices and their impact on canonical recovery probability. This prevents aggregate error rates from masking structurally harmful error modes.

9.4 Interpretation track

The interpretation track is entered only after formal and implementation tracks are stable and calibration identifiability is demonstrated.

9.4.0.1 Pre-registration and model locking.: Before hypothesis tests, the mapping form, priors, exclusion criteria, and decision thresholds are locked in a dated protocol. Post-hoc reparameterization is reported as exploratory and cannot support confirmatory claims.

9.4.0.2 Uncertainty-qualified decision rules.: All hypothesis outcomes are reported with interval estimates, posterior predictive checks, and sensitivity analysis to prior and noise-model choices. A claim is retained only when sign, scale, and direction remain stable across pre-specified robustness tests.

9.4.0.3 Layer-separation audit.: Interpretive text must explicitly cite the formal and modeling dependencies used. If any dependency fails replication, associated interpretation claims are automatically downgraded.

10 Conclusion

This manuscript proposes not only a technical formalism but also a discipline of inference. The central result is methodological: formal properties of synchronous specular rewriting can be proved and audited independently from calibration-dependent modeling statements and from cosmological interpretation hypotheses. This separation prevents common category errors in interdisciplinary work, where mathematical consistency is often overextended into empirical certainty.

At the formal level, SBA provides a local and deterministic computational substrate with exact weighted-sum preservation under Fibonacci-based rewrites and bounded commit magnitude under Protocol C arbitration. At the modeling level, dimensional bridges are treated as conditional maps that require identifiability and robustness checks. At the hypothesis level, the framework enforces pre-registration, rejection criteria, and out-of-sample replication before interpretive conclusions are considered credible.

The practical implication is that reviewers can reject one layer while preserving results in another layer, improving modular scientific progress. Future work therefore proceeds along two coupled lines: strengthening formal guarantees under broader boundary conditions and improving calibration transfer across heterogeneous substrates. If both lines mature, interpretation claims can become progressively more constrained, transparent, and testable.

Appendix A
Appendix A: Formal Assumption Ledger

This ledger records assumptions in a form suitable for line-by-line audit.

- **A1 (Finite support):** Initial non-zero occupancy is finite. Needed for well-founded termination arguments.
- **A2 (Synchronous latching):** Read snapshots are globally latched before Compute. Needed to exclude intra-round causal loops.
- **A3 (Local stencils):** Rewrites and reconstructions act on bounded neighborhoods only.
- **A4 (Deterministic tie-breaking):** Arbitration order is total, deterministic, and platform-independent.
- **A5 (Bounded commits):** Per target and round, committed value belongs to $\{-1, 0, +1\}$.

Assumptions are classified as logical (required for proof validity) or engineering (required for implementation equivalence). Any violation must be mapped to the statements it invalidates.

Appendix B
Appendix B: Reproducibility Artifact Set

A complete replication bundle must contain:

1) transition traces with round and phase timestamps,
2) declared tie-break policy and deterministic seed material,
3) formal-invariant reports and failed-case archives,
4) calibration notebooks with versioned data snapshots,
5) uncertainty and sensitivity scripts,
6) holdout and out-of-sample evaluation reports,
7) signed manifest containing hash digests for all artifacts.

Artifacts are considered valid only if hashes match and the manifest reproduces the same summary metrics when rerun in a clean environment.

Appendix C
Appendix C: Worked Case Library

The case library is organized by mechanism rather than by substrate.

- **C1: Isolated overflow.** Single $|d_i| = 2$ event, no competing targets.
- **C2: Competing-target contention.** Multiple candidates aim at one target; validates Protocol C single-winner behavior.
- **C3: Boundary-conditioned normalization.** Tests left/right edge handling under truncated neighborhoods.

- **C4: Noisy sanitization stress.** Injected ambiguity in 11-state resolution with deterministic reconstruction constraints.

Each case includes input state, expected transition trace, invariant outcomes, and admissible tolerance bands for implementation timing.

Appendix D
Appendix D: Calibration Robustness

Calibration must be stable across operating regimes rather than optimized for a single condition. The protocol includes:

1) separate fits over temperature/load/noise strata,
2) posterior-overlap diagnostics across strata,
3) drift-threshold tests for parameter transfer,
4) leave-regime-out validation to assess extrapolation risk.

A calibration is accepted only if uncertainty-adjusted predictions remain consistent under regime shifts within declared operating bounds.

Appendix E
Appendix E: Coherence Audit Rubric

A paragraph is coherent only when all items pass:

1) claim layer is explicitly labeled (formal/model/hypothesis),
2) dependencies are listed and referenceable,
3) constants are either calibrated with uncertainty or marked as uncalibrated,
4) falsification or failure conditions are stated,
5) language does not overstate epistemic status.

Audit failures are graded as minor (clarity), major (dependency omission), or critical (layer violation).

Appendix F
Appendix F: Terminology Normalization

Terminology is standardized to prevent hidden shifts in meaning.

- **Collapse**: deterministic sanitization and commit under Protocol C, not a standalone physical postulate.
- **Mass**: calibrated image of occupancy, $m_{\text{phys}} = \mu m_{\text{SBA}}$.
- **Pressure**: model-level coarse-grained load descriptor, valid only with explicit observable mapping.
- **Classicalization**: finite-round convergence to canonical bounded states under the formal semantics.

Any manuscript usage outside these meanings must declare a local redefinition.

Appendix G
Appendix G: Limits and Open Problems

The framework has known limits.

- Formal correctness does not imply physical truth.
- Proven guarantees rely on synchronous semantics; asynchronous relaxations remain partially open.
- Heterogeneous boundaries may introduce longer contention tails not yet tightly bounded.
- Calibration transfer across substrates can fail under unmodeled noise couplings.

Open problems include tight worst-case radius bounds under mixed boundary operators, probabilistic tail bounds for arbitration latency, and invariance criteria for cross-substrate calibration maps.

Appendix H
Appendix H: Extended Replication Workflow

Replication is staged to isolate failure modes:

1) **Formal parity:** reproduce invariant-preserving traces from reference inputs.
2) **Calibration parity:** reproduce parameter posteriors and predictive intervals.
3) **Hypothesis parity:** reproduce pre-registered decision outcomes.
4) **Stress-test parity:** reproduce robustness margins under controlled perturbations.

Each stage has independent pass/fail thresholds; passing a later stage never retroactively validates a failed earlier stage.

Appendix I
Appendix I: Formal Dependency Matrix

Dependencies are represented as a directed acyclic graph where nodes are statements and edges are required assumptions/lemmas. For each theorem-level claim, the matrix records:

- direct algebraic prerequisites,
- protocol assumptions,
- boundary-condition qualifiers,
- empirical inputs (if any, for modeling statements only).

This matrix enables modular critique: disputing one edge invalidates only descendants that depend on that edge.

Appendix J
Appendix J: Benchmark Design Details

Benchmark design includes both deterministic and stochastic regimes.

J.0.0.1 Deterministic regime.: Fixed seeds, fixed arbitration order, and fixed disturbance templates isolate implementation fidelity.

J.0.0.2 Stochastic regime.: Perturbations are sampled from declared distributions to estimate robustness margins.

J.0.0.3 Normalization metrics.: Cross-platform metrics are normalized by effective update count, active neighborhood width, and canonicalization depth. This avoids misleading comparisons caused by platform-specific clocking conventions.

Appendix K
Appendix K: Uncertainty Engineering

For measured ΔE_j and modeled Δm_j, infer μ in

$$\Delta E_j = \mu c^2 \Delta m_j + \epsilon_j,$$

where ϵ_j follows a declared noise model (e.g., heteroscedastic Gaussian or robust heavy-tail family).

Recommended reporting includes posterior for μ, posterior predictive checks, residual diagnostics, and sensitivity to prior width and outlier handling. Identifiability must be demonstrated by showing that μ remains bounded away from degenerate intervals under plausible model variations.

Appendix L
Appendix L: Replication Artifacts

Replicable bundles must include source hashes, trace archives, calibration notebooks, decision scripts, and manifest digests. In addition, each bundle should provide:

- environment specification (compiler/tool versions),
- deterministic seed registry,
- changelog linking manuscript claims to artifact versions,
- minimal command list for full rerun.

A claim is replication-ready only if a third party can regenerate the reported summary tables and decision outcomes from the published bundle.

Appendix M
Appendix M: Reviewer Audit Guide

Suggested review sequence:

1) identify claim layer for each major statement,
2) verify the dependency path using the formal matrix,
3) inspect calibration assumptions and uncertainty treatment,
4) check pre-registered falsification criteria,
5) confirm interpretation language does not appear inside formal proof steps.

For efficient auditing, reviewers may score each section on *layer clarity*, *dependency transparency*, and *falsifiability specificity*. Sections that fail any criterion should be revised before cross-layer conclusions are considered.

Statements and Declarations

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Competing interests: The author declares no competing interests.
Supplementary Material: Supplemental companion material is provided as `SBA_DNA.pdf`.
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